

SQL Date Functions Notes

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EXTRACT()

EXTRACT() retrieves a specific component from a date, time, or timestamp.

Instead of returning the whole date:

```
2026-07-19 14:35:28
```

it can return:

Year	Month	Day	Hour	Minute	Second
→ 2026	→ 7	→ 19	→ 14	→ 35	→ 28

Think of it as breaking a date into its individual parts.

```
SELECT *FROM OrdersWHERE EXTRACT(YEAR FROM order_date) = 2026;
```

```
SELECT EXTRACT(YEAR FROM DATE '2026-07-19');
```

```
↓  
2026
```

QUARTER

```
SELECT EXTRACT(QUARTER FROM DATE '2026-07-19');
```

```
↓  
3
```

Quarter mapping

Months	Quarter
Jan–Mar	Q1
Apr–Jun	Q2
Jul–Sep	Q3
Oct–Dec	Q4

DOW (Day of Week)

(PostgreSQL)

```
SELECT EXTRACT(DOW FROM DATE '2026-07-19');
```

Returns an integer representing the day of the week (exact numbering is vendor-specific).

9.12 DOY (Day of Year)

```
SELECT EXTRACT(DOY FROM DATE '2026-07-19');
```

↓
200

Useful for scientific calculations and analytics.

Orders in Quarter 4

```
SELECT *FROM OrdersWHERE EXTRACT(QUARTER FROM order_date)=4;
```

DATE_TRUNC()

PostgreSQL-specific.

Used to "round down" timestamps.

Syntax

```
DATE_TRUNC(unit,timestamp)
```

Truncate to Year

```
SELECT DATE_TRUNC('year',TIMESTAMP '2026-07-19 14:35:28');
```

↓

2026-01-01 00:00:00

10.4 Month

```
SELECT DATE_TRUNC('month',NOW());
```

↓

2026-07-01 00:00:00

10.5 Week

```
SELECT DATE_TRUNC('week',NOW());
```

Returns the beginning of the week according to the database's definition.

Day

```
SELECT DATE_TRUNC('day',NOW());
```

↓

2026-07-19 00:00:00

10.7 Hour

```
SELECT DATE_TRUNC('hour',NOW());
```

↓

2026-07-19 14:00:00

Formatting Dates

Converting dates into human-readable strings.

11.1 DATE_FORMAT() (MySQL)

Syntax

```
DATE_FORMAT(date, format)
```

Example

```
SELECT DATE_FORMAT('2026-07-19', '%d-%m-%Y');
```



19-07-2026

Common MySQL Format Specifiers

Specifier	Meaning
%Y	4-digit year
%y	2-digit year
%m	Month number
%d	Day
%H	24-hour
%i	Minutes
%s	Seconds
%M	Month name
%W	Weekday name

Parsing Dates

Converting strings into dates.

Formatting = Date → String

Parsing = String → Date

12.1 STR_TO_DATE() (MySQL)

```
SELECT STR_TO_DATE('19-07-2026', '%d-%m-%Y');
```



2026-07-19

CAST()

Convert compatible strings to dates.

```
SELECT CAST('2026-07-19' AS DATE);
```